Optimization of Gallium thin film deposition on elastomeric substrates

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***Abstract*— Gallium and several gallium alloys are room temperature liquid metals. Unlike mercury, it has low toxicity and no vapor pressure. Gallium has emerged as a particularly interesting material for stretchable electronics applications. We investigate physical vapor deposition (PVD) of gallium to enable production of thin film liquid metal. We consider two factors: the total mass of deposited gallium and the adhesion layer required for the deposition. The response parameter for our model is the roughness of the deposited gallium layer. We find that the roughness of the adhesion layer has a first order effect.**

I. Introduction

Stretchable electronics consists in incorporating electronic functions such as interconnections, electrodes and transducers on soft, elastomer substrates. Applications include skin-like artificial sensing systems [1] and ultra-compliant implants for neuro-stimulation and recording [2]. Liquid metals have been extensively studied to form soft electrical connections and sensors, since they can maintain metallic conduction under very large strains (up to 800 %). Reported deposition and patterning techniques include injection into micro-channels, stamping, screen-printing, spraying, laser patterning, direct writing and micro-contact printing [3]. However, these techniques present limitations in terms of minimum feature sizes or scalability to large area patterning with high throughput.

We are investigating ways to form continuous thin films of gallium (Ga) using physical vapor deposition (PVD). Gallium is a metal that melts at 30 °C and has a tendency to supercool, remaining liquid at room-temperature once melted. In our current process, a silicone (polydimethylsiloxane, PDMS) substrate is coated with a thin film (typically 40 nm) of gold. A 5 nm adhesion layer of titanium or chromium can be deposited prior to gold coating. Then, a known mass of gallium (typically 0.25 g) is thermally evaporated and forms a continuous film with an average thickness of the order several hundreds of nanometers. The film maintains electrical conductivity even when the substrate is stretched up to 100 %. Our process is compatible with deposition over large areas in a parallel fashion and opens up micro-patterning techniques, such as photolithography, thanks to the thin nature of the deposited liquid metal film.

Observation of the surface of the gallium film revealed a high surface roughness, with accumulation of gallium in the form of droplets about 10 μm high (Fig. 1). Such high roughness and local accumulation of gallium is detrimental to the electro-mechanical properties of the film, since area with less metal is more prone to crack and disrupt conductivity of the film. In fact, we observed that high roughness correlated with higher initial resistivity and higher sensitivity to large mechanical strain for the deposited films.

Two consecutive sets of experiments were performed. In a first phase we considered two factors, the mass of gallium and the adhesion layer with two discrete conditions (5/20 nm Ti-Au or 5/20 nm Cr-Au). The nature of the adhesion layer was expected to have little influence on the roughness of the deposited layer.
A second set of experiments was then scheduled to explore in more detail the effect of the roughness of the adhesion layer. The discrete adhesion layer factor was transformed in a continuous factor by measuring the roughness of the adhesion layer film prior to the gallium deposition. The analysis of the design of experiment and the results are discussed in the following sections. The experimental mind map is given in figure 2.

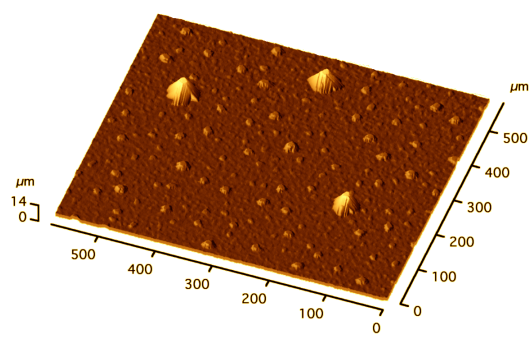

Fig. 1. Optical profiler image of a gallium thin film on a PDMS substrate coated with 40 nm of Au prior to gallium deposition. Evaporated mass is 0.25g.

II. First set of experiments

A. Experimental procedure

A 125 μm layer of silicone (polydimethylsiloxane, PDMS) is deposited on a 100 mm silicon wafer by spin-coating and cured in an oven. The PDMS layer is then further coated with a thin film of chromium-gold (5/20 nm)or titanium-gold (5/20 nm). Finally, a gallium film is deposited via physical vapor deposition (PVD) technique. Up to three wafers can be processed at the same time in the PVD machine used for gallium deposition.

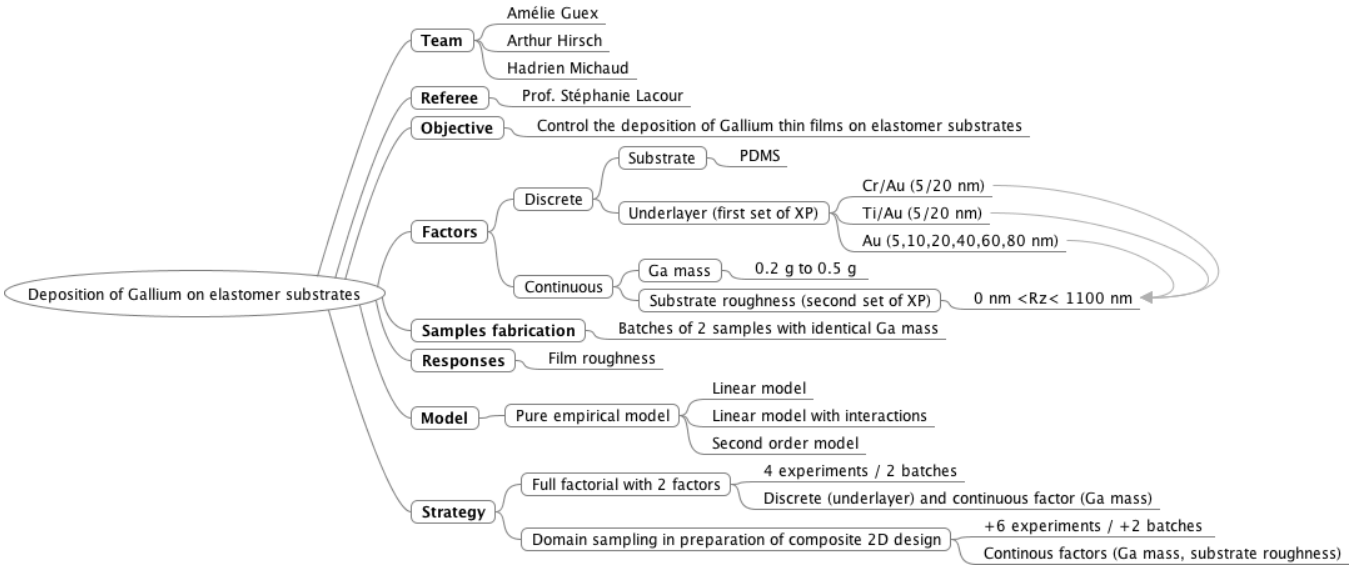

Fig. 2. Experimental Mind Map

After deposition, the roughness of the deposited film is measured using an optical profilometer (VEECO WYKO NT1100) on a 300μm x 300μm evaluation area (vertical resolution $<$ 1 nm, spatial resolution $<$ 1 μm).

B. Factors

We varied two factors during the first stage of our study:

- **the total mass of gallium** deposited on the substrates could be varied between 0.2 g and 0.5 g. This factor was continuous.
- **the nature of the coating film** could be varied between Ti/Au and Cr/Au. This factor was discrete in the first set of experiment.

C. Responses

We chose as response the gallium film roughness measured with the following parameter:

- **Rz** : The Average maximum profile of the ten greatest peak-to-valley separations in the evaluation area.

D. Design of experiments

1) Model choice: A linear, linear with interactions and second degree models will be considered and compared.

The equations of the models are given by the following formulas :

$$Y_1 = \alpha_0 + \alpha_1 * x_1 + \alpha_2 * x_2 \quad (1)$$

$$Y_2 = \alpha_0 + \alpha_1 * x_1 + \alpha_2 * x_2 + \alpha_{12} * x_1 * x_2 \quad (2)$$

$$Y_3 = \alpha_0 + \alpha_1 * x_1 + \alpha_2 * x2 + \alpha_{12} * x_1 * x_2 + \alpha_{11} * x_1^2 + \alpha_{22} * x_2^2 \quad (3)$$

2) Matrix of experiment: A factorial design was chosen for the first set of experiments. Two factors were considered, the mass of gallium and the underlayer type. For each gallium mass and underlayer type, four values were obtained, for a total of 16 experiments. For this design, only the linear and linear with interactions designs can be considered. Figure 3 shows the chosen experimental domain.

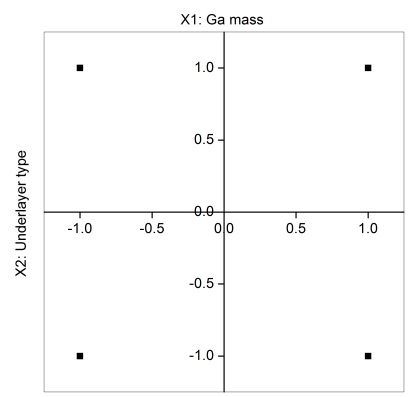

Fig. 3. Experimental domain for the first phase of experiments

After calculation of the dispersion and correlation matrix, the variance inflation factors and their associated functions can be computed. The factorial design enables to have value of the inflation factors equal to one. Figure 4 shows the associated variance functions for both linear and linear with interaction models. The variance functions have values below one which should guarantee a low variance of the coefficient estimation.

E. Results

After performing the experiments, the coefficients of the model were computed using least square fit. Table I shows

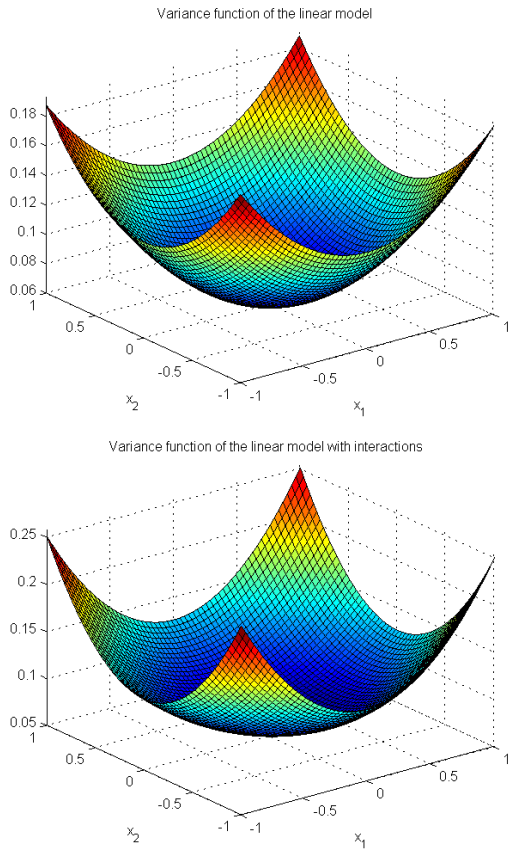

Fig. 4. Variance function plot for linear and linear with interactions models for the first phase of experiments

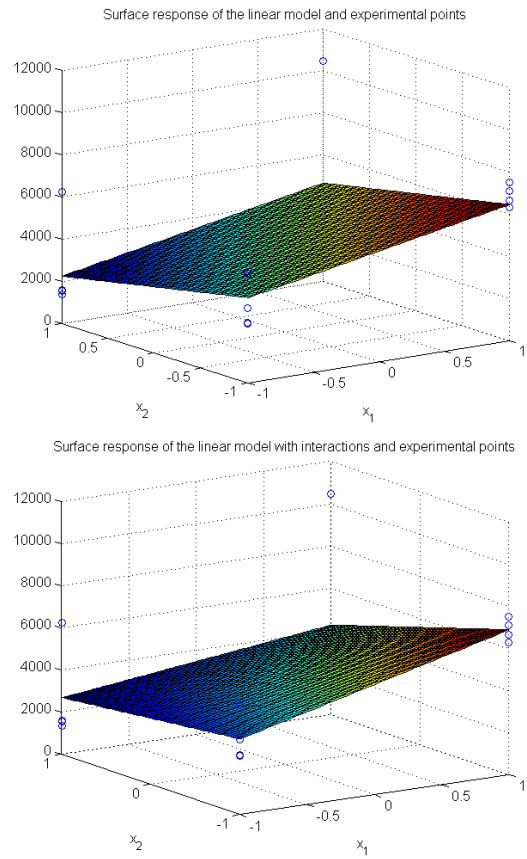

Fig. 5. Linear regression plots for linear and linear with interactions models (phase1).

the linear regression coefficients for the linear and linear with interactions models. Due to the orthogonality of the factors, first degree coefficients are similar for both models. Figure 5 gives the linear regression plots for both models.

TABLE I

LINEAR REGRESSION COEFFICIENTS FOR THE LINEAR AND LINEAR WITH INTERACTIONS (PHASE 1)

	Linear	Linear and interactions
$\alpha 0$	4352.5	4352.5
$\alpha 1$	1211.25	1211.25
$\alpha 2$	-890	-890
$\alpha 12$		-433.75

We performed an Anova analysis in order to verify the statistical significance of the models and factors. Table II shows the anova results for the models and table III for the factors.

The p-value of the linear model shows statistical significance for the linear model. However, adding the interactions did not show a statistical significance.

From table III we can see that the most important effect is the constant (α_0) and the mass of Gallium (α_1). For α_0 and α_1 the p-value is below 0.05 and thus statistically significant.

TABLE II

ANOVA OF THE MODELS (PHASE1)

	SS	df	ms	F	p
linear	3.392E08	3	1.131E08	18.462	8.607E-05
interactions	3.010E06	1	3.010E06	0.491	0.497
residue	7.350E07	12	6.125E06		
measurements	4.158E08	16			

On the other hand the effect of interaction between the two factors is not significant.

F. Conclusion

With this first experimental design, the mass of gallium appears to be significant as opposed to the underlayer type. The linear model thus satisfactorily describes the data.

III. SECOND SET OF EXPERIMENTS

After the first phase of experiments, the discrete factor (nature of the coating film) was modified into a continuous factor (roughness of the underlayer). This allows a better characterization of the effect. Precedent experiments could be included by measuring the roughness of the previously used adhesion layers. When deposited the gold adhesion layer buckles and creates a rough surface: the thicker the gold layer the rougher the surface. Thicknesses of 5, 10, 20, 40, 60 and

TABLE III

ANOVA OF THE FACTORS FOR THE LINEAR MODEL (PHASE 1)

	SS	df	ms	F	p
α_0	3.03E+08	1	3.03E+08	19.21	7.37E-07
α_1	2.34E+07	1	2.34E+07	3.99	0.03
α_2	1.27E+07	1	1.27E+07	2.15	0.09
residue	1.62E+08	13	5.89E+06		
total	4.15E+08	16			

80 nm of Au were selected. Theses thicknesses allowed to test a range of roughnesses, even though the roughness is not linearly linked to the thickness of deposited gold. Four measurements per condition were performed.

A. Experimental procedure

The experimental procedure was similar to the first phase but before deposition, the roughness of the deposited adhesion film is measured using an optical profilometer (VEECO WYKO NT1100) on a 100μm x 100μm evaluation area (vertical resolution $<$ 1 nm, spatial resolution $<$ 1 μm).

B. Factors

We varied two factors during the second stage of our study:

- **the total mass of gallium** deposited on the substrates could be varied between 0.2 g and 0.5 g. This factor was continuous.
- **the roughness of the adhesion layer Rz** : The Average maximum profile of the ten greatest peak-to-valley separations in the evaluation area. This factor is continuous contrary to the first phase of experiments.

C. Responses

The analyzed response in this second phase of experiments is similar to the first phase, the roughness of the gallium film.

D. Design of experiments

1) Model choice: As in the first phase linear, linear with interactions and second degree models will be considered and compared.

2) Matrix of experiments: The experimental domain for the second set of experiments is given in figure 6. The design of experiment is non orthogonal for both linear and second order models as shown in figure 7 where the angles between estimators are computed.

The dispersion matrix for the different models are given in tables IV, V and VI. The diagonal elements of these matrix should remain as small as possible in order not to minimize amplification of the variance from the factors to the coefficients. For the linear model the diagonal elements are smaller than one. However it is not the case for some diagonal elements of the linear model with interactions and the second order model. These two last models are thus more prone to amplify the variance of the factors.

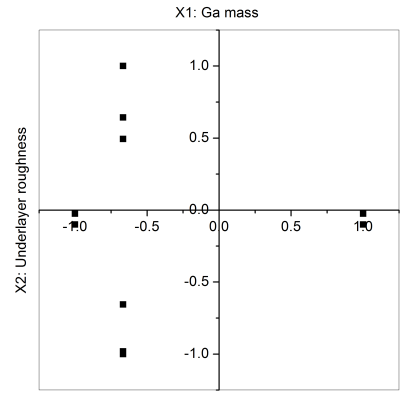

Fig. 6. Experimental domain for the second phase of experiments

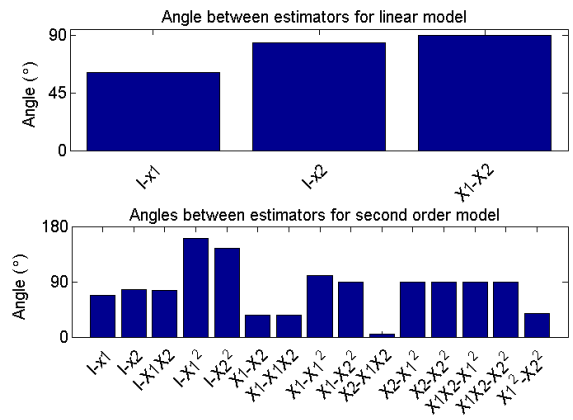

Fig. 7. Angles between estimators for linear and second order model with the second phase design.

TABLE IV

DISPERSION MATRIX OF THE LINEAR MODEL (PHASE 2)

	Cst	x1	x2
Cst	0.033	0.0197	0.0045
x1	0.0197	0.0493	-0.0004
x2	0.0045	-0.0004	0.0623

TABLE V

DISPERSION MATRIX OF THE LINEAR MODEL WITH INTERACTIONS (PHASE 2)

	Cst	x1	x2	x1*x2
Cst	0.0878	0.1017	0.8714	1.3022
x1	0.1017	0.1726	1.3022	1.9566
x2	0.8714	1.3022	13.8322	20.6836
x1*x2	1.3022	1.9566	20.6836	31.0685

TABLE VI

DISPERSION MATRIX OF THE SECOND DEGREE MODEL (PHASE 2)

	Cst	x1	x2	x1*x2	x1*x1	x2*x2
Cst	0.9766	0.1569	0.8037	1.3014	-0.9730	-0.5479
x1	0.1569	0.1857	1.3025	1.9564	-0.0749	1.44E-06
x2	0.8037	1.3025	13.8394	20.6836	0.0675	0.0576
x1*x2	1.3014	1.9564	20.6836	31.0685	0.0011	2.29E-05
x1*x1	-0.9730	-0.0749	0.0675	0.0011	1.0868	0.5491
x2*x2	-0.5479	1.44E-06	0.0576	2.29E-05	0.5491	0.4573

Similarly, the variation inflation factors given in Table VII show that only the linear model is valid in this situation. Indeed the linear with interactions and second degree models have values higher than 10.

TABLE VII

VIF OF THE LINEAR, LINEAR WITH INTERACTIONS AND SECOND DEGREE MODELS (PHASE 2)

Cst	1.329	3.5121	39.0625
x1	1.3159	4.6017	4.9519
x2	1.0142	225.2511	225.3692
x1*x2		226.3407	226.3408
x1*x1			22.5416
x2*x2			6.1168

Variance functions are given in figure 8. If the variance function is satisfying for the whole domain in the case of the linear model, it is not the case for the two others models. There is a clear inflation of variance when increasing the mass of gallium.

E. Results

After performing the experiments, the coefficients of the model were computed using least square fit. Coefficients for the linear, linear with interactions and second degree models are shown in table VIII and the regression plots are given in figure 9. As the factors are not orthogonal, the first degree coefficients vary with the different models.

TABLE VIII

LINEAR REGRESSION COEFFICIENTS FOR THE LINEAR, LINEAR WITH INTERACTIONS AND SECOND DEGREE MODELS (PHASE 2)

α_0	5423.8678	4653.9526	11559.2235
α_1	434.4343	-722.4038	53.8174
α_2	-1490.4743	-13719.7392	-14085.6187
$\alpha_1 2$		-18369.401	-18380.5968
$\alpha_1 1$			-8077.4177
$\alpha_2 2$			-3038.5192

The anova analysis of the model is given in table IX. It shows that only the linear and second degree models are statistically significant. For this reason, only these two models will be considered for the rest of the analysis.

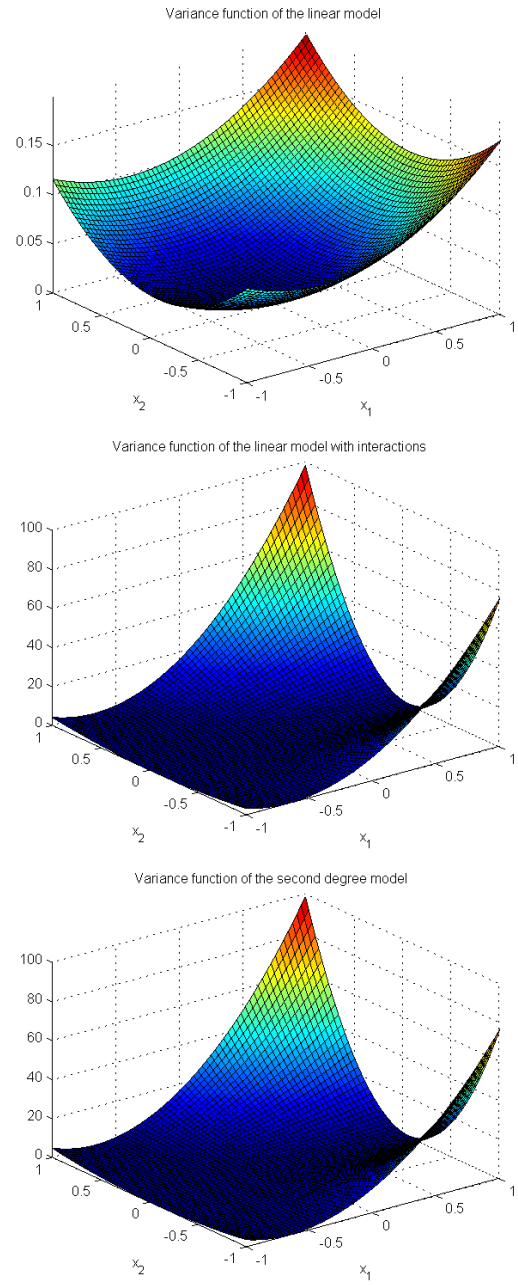

Fig. 8. Variance function plot for linear, linear with interactions and second degree models for the second phase of experiments (phase 2)

The design is non-orthogonal for linear and second order model (see Fig. 7). We calculated the corrected sum of squares for both models in order to identify statistically significant effects.

1) Linear model: Influence of the substrate roughness is statistically significant with a p-value < 0.05 while placed in first position of the corrected sum of square. Influence of the gallium is not, with a p-value > 0.05 even if placed last in the order for orthogonalization.

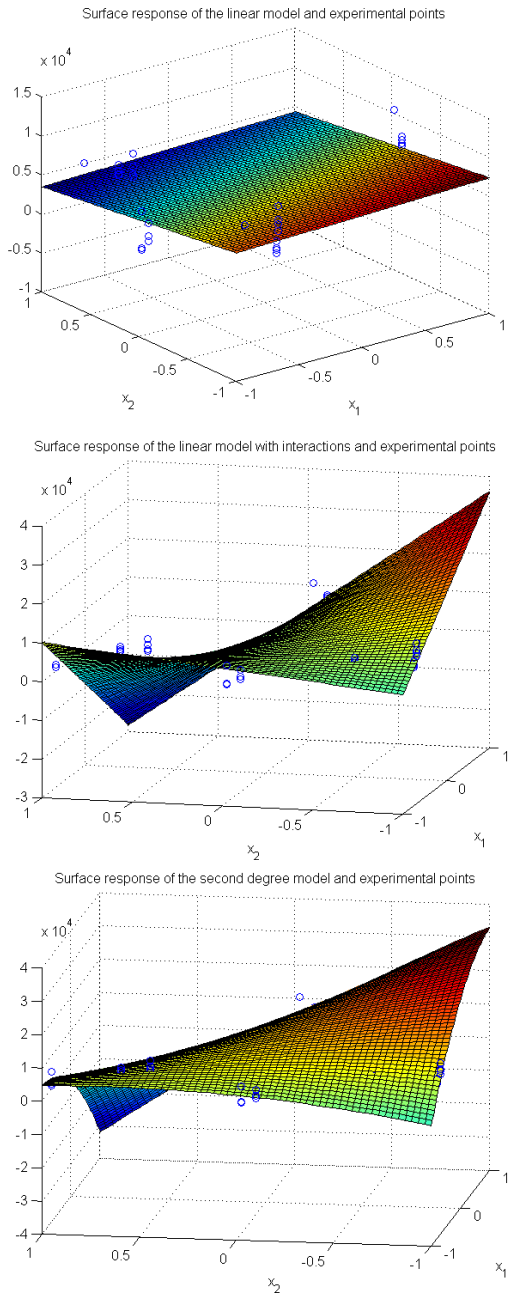

Fig. 9. Linear regression plots for linear, linear with interactions and second degree models (phase 2).

TABLE IX

ANOVA OF THE MODELS (PHASE 2)

	SS	df	ms	F	p
linear	1.19E+09	3	3.97E+08	83.2	9.75E-16
interactions	1.00E+07	1	1.00E+07	2.1	0.157
2nd degree	6.60E+07	2	3.30E+07	6.93	0.003
residue	1.62E+08	34	4.76E+06		
measurements	1.43E+09	40			

TABLE X

ANOVA OF THE FACTORS FOR THE LINEAR MODEL WITH CORRECTED SUM OF SQUARES (PHASE 2)

	SS	SS*	DF	MS	F	p
α_0		1.2×10^9	1	1.2×10^9	178	1.0×10^{-15}
α_2		3.6×10^7	1	3.6×10^7	5.5	0.02
α_1		3.8×10^6	1	3.8×10^6	0.6	0.44
residue	2.4×10^8	2.4×10^8	37	6.4×10^8	1	
Total		1.4×10^9	40			

2) Second order model: We test different sequences to calculate the corrected sum of squares associated with the second order model. Since the model is non-orthogonal, the p-values depend on the order in which the coefficient are considered. In order for an effect to be significant, its p-value must be lower than 0.05 even when considered first. The first sequence discards a first order effect for factor 1 (p-value associated to $\alpha_1 > 0.05$), second order effect for factor 2 (p-value associated to $\alpha_{22} > 0.05$) and interaction between factor 1 and 2 (p-value associated to $\alpha_{12} > 0.05$). Sequence 2 confirms first order effect for factor 2 (p-value associated to $\alpha_2 < 0.05$). Sequence 3 confirms second order effect for factor 1 (p-value associated to $\alpha_{11} < 0.05$). In conclusion, only factor 2 (substrate roughness) has a significant first order effect and only factor 1 (gallium mass) has a significant second order effect.

TABLE XI

P-VALUES FOR VARIOUS ORDERS OF COEFFICIENTS, WITH VARIOUS ORDERS FOR ORTHOGONALIZATION.

	p		p		p
α_0	5×10^{-17}	α_0	5×10^{-17}	α_0	5×10^{-17}
α_2	0.01	α_2	0.01	α_{11}	0.02
α_1	0.38	α_{11}	0.02	α_2	0.01
α_{22}	0.27	α_1	0.03	α_1	0.03
α_{11}	0.001	α_{22}	0.05	α_{22}	0.05
α_{12}	0.14	α_{12}	0.14	α_{12}	0.14

F. Conclusion

Statistical analysis of the results of the second set of experiments reveals that substrate roughness has a significant first order effect for linear and second order models, while gallium mass has a second order effect for a second order model.

IV. Discussion

The design of experiments for phase 2 is not very satisfying. The analysis of the matrix of experiment given in section III-D.2 showed that only the linear model had reasonable VIFs. In order to consider second order model or model with interactions, extra experimental points would be required. We propose to add two points at 0.5g of gallium and 10/60 nm of Au for adhesion layer. The resulting experimental domain is shown in figure 10.

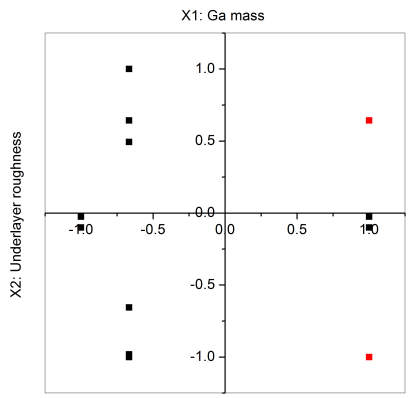

Fig. 10. Experimental domain with two additional datapoints for next phase

The new VIFs values can then be calculated and are given in table XII. Both linear and linear with interactions models would then have VIFs close to one. However the VIFs of the second order models will still be large.

To have a a reliable second order models extra points would be required in the center of the domain. This would give us a design close to the composite 2D design.

TABLE XII

VIF of the linear, linear with interactions and second degree models with the two extra experimental points

Cst	1.061	1.068	31.995
x1	1.041	1.076	1.862
x2	1.020	1.135	1.460
x1*x2		1.142	1.241
x1*x1			21.064
x2*x2			4.617

V. Conclusions

The course of design of experiment allowed us to build experimental designs and analyze their limitations in term of statistical significance. In the first phase of experiments, the adhesion layer was considered as a discrete factor and was found not to have a significant effect compared to the gallium mass. However in the second phase the adhesion factor was related to the roughness of the layer thus providing a continuous factor. In this case the roughness was found to have a higher effect than the mass of gallium. However the chosen design allowed us only to consider linear model. In the future we would like to add additional data points in the domain in order to allow the evaluation of a model with interactions and a second order model.

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